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COMPREHENSIVE SOLAR PV, BESS & ENERGY SYSTEMS PROJECT REPORT

Master Engineering Dossier — 17 Field Installations & Industrial Telemetry Deployments

DATE: OCTOBER 2026

TOTAL SOLAR PV	UTILITY HIGH-VOLTAGE	COMMISSIONED SITES	BESS STORAGE DEPLOYED
1.63 MWp	1,500 VDC Bus	17 Projects	500+ kWh

1. MASTER PROJECT SUMMARY MATRIX

#	Project Title	Client	Category	PV Array	Inverter AC	Storage (BESS)	Status
01	**1.3 MWp Rooftop High-Voltage Grid-Connected Solar PV at The Link**	The Link, Tatu City	Utility Grid-Tied	1.30 MWp DC	1.11 MW AC	None	Completed
02	**Mushroom Motors Commercial Microgrid & Solar Carport**	Mushroom Motors	C&I Solar & BESS	40.04 kWp DC	36 kW AC	48 kWh LiFePO4	Completed
03	**Prof. Jacob Bifacial Carport Microgrid**	Prof. Jacob (Karen)	Residential Hybrid	12.78 kWp DC	15 kW AC	32 kWh LiFePO4	Completed
04	**Runda Bifacial Solar Carport & Energy Storage System**	Runda Residence	Residential Hybrid	12.78 kWp DC	36 kW AC	32 kWh LiFePO4	Completed
05	**Kiambu Water Pump Solar Hybrid Plant**	Kiambu Water	C&I Solar & BESS	30.03 kWp DC	45 kW AC	48 kWh LiFePO4	Completed
06	**Mombasa 3-Phase Residential Hybrid System**	Mombasa Residence	Residential Hybrid	10.92 kWp DC	18 kW AC	30 kWh LiFePO4	Completed
07	**Carol Daykio Residence Hybrid Solar & Storage**	Carol Daykio Residence	Residential Hybrid	7.28 kWp DC	12 kW AC	16 kWh LiFePO4	Completed
08	**Mrs. Catherine Garden City Residence Hybrid Solar & Storage**	Mrs. Catherine	Residential Hybrid	7.28 kWp DC	12 kW AC	16 kWh LiFePO4	Completed
09	**BM Security Operations Center Solar Backup**	BM Security	C&I Solar & BESS	3.64 kWp DC	6 kW AC	10 kWh LiFePO4 Battery	Completed
10	**Mariakani Coastal Residential Hybrid PV System**	Private Residence	Residential Hybrid	12.18 kWp DC	12 kW AC	32 kWh LiFePO4	Completed
11	**Willis Auto Spa Solar PV & Pumping Energy System**	Willis Auto Spa	C&I Solar & BESS	15.47 kWp DC	36 kW AC	32 kWh LiFePO4	Completed
12	**Oaklands 5 Villas Residential Hybrid System**	Oaklands 5 Villas	Residential Hybrid	4.55 kWp DC	6 kW AC	16 kWh LiFePO4	Completed
13	**Power Analysis & Load Calculations**	Residential & Industrial	IoT & Automation	Demand Profiling & Logging	Capacity Sizing Models	Phase Balancing & Harmonics	Continuous
14	**Tsuku Tsuku Hospitality Solar & Storage System**	Tsuku Tsuku	C&I Solar & BESS	30.03 kWp DC	36 kW AC	96 kWh LiFePO4	Completed
15	**ESP32 Cloud-Connected EV Charge-Point Status Reporter**	Engineering R&D Infrastructure	IoT & Automation	Station Power Coupling Subsystem	AC & DC EV Fast-Charger Coupling	Bidirectional MQTT Cloud Telemetry & NVS	Completed
16	**Rainbow Resort Commercial Hybrid Solar PV & High-Capacity BESS**	Rainbow Resort	C&I Solar & BESS	145.60 kWp DC	150 kW AC	336 kWh LiFePO4	Completed
17	**Industry 4.0 Real-Time Energy Telemetry & Cloud Analytics**	Digital Qatalyst Projects	IoT & Automation	Sub-Station Meter Network & Telemetry Ingestion	Multi-Point Power Monitoring	Real-Time Cloud Dashboards & Analytics	Completed

2. DETAILED TECHNICAL SPECIFICATIONS & INSTALLATION RECORDS

01. 1.3 MWp Rooftop High-Voltage Grid-Connected Solar PV at The Link

UTILITY GRID-TIED | COMPLETED

Client: The Link, Tatu City | Location: Tatu City, Kiambu County, Kenya | Period: December 2025 – March 2026 | Contractor: Hounen & Keda Solar

PV ARRAY (DC)	INVERTER (AC)	STORAGE (BESS)	INVERTER BRAND
1.30 MWp DC	1.11 MW AC	None	Huawei

Engineering Summary:
The 1.3 MWp rooftop solar installation at The Link in Tatu City delivers 1.11 MW AC synchronized power to solve three critical challenges for local businesses: bypassing expensive national grid retail tariffs for SMEs, shielding logistics and manufacturing firms from grid unreliability and blackout downtime, and replacing fossil-fuel energy to meet international sustainability standards.

Operational Challenge: Businesses and SMEs operating at The Link in Tatu City faced three major problems: steep operational costs driven by expensive national grid retail tariffs, chronic national grid unreliability causing costly downtime for logistics and manufacturing facilities during blackouts, and fossil-fuel reliance creating barriers to meeting strict international carbon-reduction and eco-certification standards.

Engineering Solution: Engineered a 1.30 MWp rooftop solar plant comprising 2,052 high-density 635 W bifacial modules across a 1,500 VDC bus topology, synchronized into medium-voltage distribution infrastructure via six 185 kW Huawei high-voltage string inverters to provide cheap, clean, and reliable daytime power.

Key Technical Highlights:
- Bypasses expensive national grid retail tariffs, providing SMEs with cheap, predictable electricity rates
- Protects logistics and manufacturing companies from costly downtime caused by national grid blackouts during peak daylight hours
- Replaces fossil-fuel-reliant energy with clean solar power to help firms meet strict international carbon-reduction and eco-certifications

Technologies: Huawei 185 kW Inverters • 1,500 VDC Bus Topology • Underground Raceways & Trays • MV Transformer Coupling • SCADA Monitoring Nodes • IV-Curve Tracing

02. Mushroom Motors Commercial Microgrid & Solar Carport

C&I SOLAR & BESS | COMPLETED

Client: Mushroom Motors | Location: Kiambu County, Kenya | Period: May – June 2026 | Contractor: Voltifix New Energy SEZ Limited

PV ARRAY (DC)	INVERTER (AC)	STORAGE (BESS)	INVERTER BRAND
40.04 kWp DC	36 kW AC	48 kWh LiFePO4	Growatt

Engineering Summary:
Turnkey 40.04 kWp solar carport canopy and workshop microgrid designed to drastically reduce overall utility power costs and primarily power the business centre and automotive workshops using clean solar energy, supported by a 48 kWh LiFePO4 battery bank and Growatt parallel hybrid inverters to minimize grid outage disruptions.

Operational Challenge: Escalating utility grid electricity tariffs and recurrent grid blackouts severely impacted operations across the business centre and automotive workshops, resulting in high recurring operational power bills and disruptive downtime for diagnostic equipment and vehicle hoists.

Engineering Solution: Engineered a 40.04 kWp structural steel solar carport canopy to primarily power the business centre and automotive facility using solar energy, coupled with three 12 kW Growatt parallel hybrid inverters (36 kW AC), a 48 kWh LiFePO4 battery bank, and an ATS cabinet with inductive surge suppression.

Key Technical Highlights:
- Primarily powers the business centre and workshops with solar energy to drastically slash utility grid power costs
- Minimizes grid outage downtime with 48 kWh LiFePO4 battery storage and automated transfer switching
- Dual-purpose structural steel carport providing customer vehicle shade and 40.04 kWp clean generation

Technologies: Growatt Hybrid Inverters • LiFePO4 Server-Rack Batteries • Solar Carport Racking • CAN/RS485 Bus • Automatic Transfer Switch (ATS) • Motor Inrush Protection

03. Prof. Jacob Bifacial Carport Microgrid

RESIDENTIAL HYBRID | COMPLETED

Client: Prof. Jacob (Karen) | Location: Karen, Nairobi, Kenya | Period: September 2026 | Contractor: Voltifix New Energy SEZ Limited

PV ARRAY (DC)	INVERTER (AC)	STORAGE (BESS)	INVERTER BRAND
12.78 kWp DC	15 kW AC	32 kWh LiFePO4	Growatt

Engineering Summary:
Designed and implemented a 12.78 kWp residential bifacial carport microgrid in Karen, engineered to reduce overall power costs from the utility grid and eliminate household power shortages during grid disturbances through a 15 kW Growatt hybrid inverter, 32 kWh LiFePO4 battery bank, and intelligent sub-panel segregation.

Operational Challenge: Escalating utility grid electricity tariffs and recurrent grid disturbances in Karen subjected the residence to frequent power shortages, while heavy domestic loads generated steep recurring monthly grid power bills.

Engineering Solution: Erected a custom carport array of 18 units of 710 W bifacial panels (12.78 kWp) to maximize solar self-generation, integrated a single 15 kW Growatt hybrid inverter with 32 kWh LiFePO4 storage, and physically segregated essential and non-essential sub-panels.

Key Technical Highlights:
- Substantially cuts utility grid power costs by maximizing bifacial solar self-consumption
- Minimizes power shortages during grid disturbances via 32 kWh LiFePO4 storage and automated failover
- 18 units of 710 W bifacial modules mounted on custom steel carport framing

Technologies: 710W Bifacial Modules • Growatt 15 kW Hybrid Inverter • 32 kWh LiFePO4 Storage • Parallel Busways • Sub-Panel Segregation • Automated Grid Failover

04. Runda Bifacial Solar Carport & Energy Storage System

RESIDENTIAL HYBRID | COMPLETED

Client: Runda Residence | Location: Runda, Nairobi, Kenya | Period: July 2026 | Contractor: Voltifix New Energy SEZ Limited

PV ARRAY (DC)	INVERTER (AC)	STORAGE (BESS)	INVERTER BRAND
12.78 kWp DC	36 kW AC	32 kWh LiFePO4	Jua Power

Engineering Summary:
Engineered an architectural 12.78 kWp bifacial solar carport paired with three 12 kW Jua Power hybrid inverters in 3-phase parallel and 32 kWh LiFePO4 storage, built to reduce overall power costs from the utility grid and eliminate power shortages during grid disturbances for luxury residential loads and EV charging.

Operational Challenge: Escalating utility grid power tariffs and frequent grid disturbances in Runda caused recurring power shortages for heavy air conditioning, borehole pumps, and EV charging, while delicate clay roof tiles precluded roof mounting.

Engineering Solution: Constructed an architectural steel carport fitted with 18 large-format 710 W bifacial dual-glass panels (12.78 kWp) to generate primary solar power, paired with three 12 kW Jua Power 3-phase parallel inverters and 32 kWh LiFePO4 storage.

Key Technical Highlights:

- Drastically cuts overall utility power costs by supplying luxury household loads via high-output solar
- Shields the residence from grid disturbances and power shortages with 32 kWh LiFePO4 storage
- High-density 710 W bifacial dual-glass modules harvesting ground albedo reflection

Technologies: 710W Bifacial Dual-Glass Panels • Jua Power 12 kW Inverters • LiFePO4 Storage • Architectural Carport • EV Charger Integration • Smart Charging Profiles

05. Kiambu Water Pump Solar Hybrid Plant

C&I SOLAR & BESS | COMPLETED

Client: Kiambu Water | Location: Kiambu County, Kenya | Period: August 2026 | Contractor: Voltifix New Energy SEZ Limited

PV ARRAY (DC)	INVERTER (AC)	STORAGE (BESS)	INVERTER BRAND
30.03 kWp DC	45 kW AC	48 kWh LiFePO4	Growatt

Engineering Summary:
Engineered a 30.03 kWp elevated solar canopy installation with 45 kW AC continuous output and 48 kWh LiFePO4 battery storage, built to primarily power the borehole pump using solar energy, reduce overreliance on the utility grid, and substantially reduce overall electricity costs from the grid.

Operational Challenge: Heavy overreliance on the utility grid for continuous borehole and booster water pumping resulted in high recurring electricity bills, while frequent grid disturbances and outages caused disruptive water supply shortages and pump downtime.

Engineering Solution: Installed a 30.03 kWp elevated steel canopy solar array engineered to primarily power the borehole pump using solar energy, backed by three 15 kW Growatt hybrid inverters in 3-phase parallel (45 kW AC), a 48 kWh LiFePO4 battery bank, and Variable Frequency Drive (VFD) motor controls.

Key Technical Highlights:

- Primarily powers the borehole water pump using solar energy to minimize utility grid overreliance and slash power costs
- Delivers balanced 415V 3-phase power tailored for high-inductance pumping motor startup
- 48 kWh battery storage guarantees uninterrupted water supply during utility grid load-shedding

Technologies: Growatt 15 kW Inverters • LiFePO4 Battery Bank • Variable Frequency Drive (VFD) • Motor Start Suppression • Class-II Surge Arrestors • Phase Rotation Verification

06. Mombasa 3-Phase Residential Hybrid System

RESIDENTIAL HYBRID | COMPLETED

Client: Mombasa Residence | Location: Mombasa, Kenya | Period: September 2026 | Contractor: Voltifix New Energy SEZ Limited

PV ARRAY (DC)	INVERTER (AC)	STORAGE (BESS)	INVERTER BRAND
10.92 kWp DC	18 kW AC	30 kWh LiFePO4	Jua Power

Engineering Summary:
Installed a 10.92 kWp 3-phase residential hybrid solar and 30 kWh LiFePO4 battery system in Mombasa, engineered to reduce overall utility grid electricity costs and minimize power shortages during coastal grid disturbances while maintaining balanced power for continuous air conditioning.

Operational Challenge: Steep utility grid power costs combined with chronic coastal grid disturbances and blackouts led to severe household power shortages, leaving the residence without reliable electricity to run essential air conditioning and domestic appliances during humid heatwaves.

Engineering Solution: Engineered a balanced 3-phase system with 10.92 kWp rooftop solar, three 6 kW Jua Power hybrid inverters synchronized across L1/L2/L3, and 30 kWh of LiFePO4 battery storage configured to maximize self-consumption and sustain critical cooling loads.

Key Technical Highlights:

- Substantially cuts monthly utility grid power costs through high-yield rooftop solar generation
- Minimizes power shortages during coastal grid disturbances with 30 kWh LiFePO4 battery storage
- Balanced 3-phase architecture (L1, L2, L3) prevents neutral overheating and phase drift

Technologies: Jua Power 6 kW Inverters • 3-Phase Balancing • 30 kWh LiFePO4 Storage • Anti-Corrosion Mounting • Air Conditioning Tuning • Low-Voltage DC Bus

07. Carol Daykio Residence Hybrid Solar & Storage

RESIDENTIAL HYBRID | COMPLETED

Client: Carol Daykio Residence | Location: Daykio, Kiambu County, Kenya | Period: September 2026 | Contractor: Voltifix New Energy SEZ Limited

PV ARRAY (DC)	INVERTER (AC)	STORAGE (BESS)	INVERTER BRAND
7.28 kWp DC	12 kW AC	16 kWh LiFePO4	Jua Power

Engineering Summary:
Engineered a 7.28 kWp residential hybrid solar PV and 16 kWh LiFePO4 battery installation on a tile roof at Daykio, designed to reduce overall power costs from the utility grid and eliminate household power shortages during grid disturbances for uninterrupted living and remote work.

Operational Challenge: High electricity bills from the utility grid and unpredictable grid disturbances resulted in recurring power shortages that interrupted remote work and household living, while delicate clay roof tiles demanded non-penetrating mounting to prevent water ingress.

Engineering Solution: Installed a 7.28 kWp solar PV array using non-penetrating stainless tile brackets and weatherproof flashings, coupled to a 12 kW Jua Power hybrid inverter, dual standalone LiFePO4 batteries (16 kWh), and smart energy management to prioritize self-generation.

Key Technical Highlights:
- Significantly reduces overall utility grid electricity expenses through high-efficiency rooftop generation
- Eliminates household power shortages during grid disturbances via 16 kWh LiFePO4 battery storage
- Specialized residential tile roof flashing preventing water ingress

Technologies: Jua Power 12 kW Inverter • LiFePO4 Standalone Batteries • Tile Roof Mounting Brackets • DC Isolators • Wi-Fi Telemetry Dongle • Mobile App Monitoring

08. Mrs. Catherine Garden City Residence Hybrid Solar & Storage

RESIDENTIAL HYBRID | COMPLETED

Client: Mrs. Catherine | Location: Garden City, Nairobi, Kenya | Period: October 2026 | Contractor: Voltifix New Energy SEZ Limited

PV ARRAY (DC)	INVERTER (AC)	STORAGE (BESS)	INVERTER BRAND
7.28 kWp DC	12 kW AC	16 kWh LiFePO4	Jua Power

Engineering Summary:
Engineered a 7.28 kWp residential hybrid solar PV and 16 kWh LiFePO4 battery installation for Mrs. Catherine in Garden City, Nairobi. Designed to substantially reduce grid electricity expenditure and ensure seamless domestic power continuity during urban utility outages.

Operational Challenge: High utility electricity bills and frequent urban power outages in Garden City disrupted household operations, requiring a reliable, high-capacity hybrid solar and battery storage system with zero-delay failover.

Engineering Solution: Installed a 7.28 kWp rooftop array consisting of 16 high-efficiency 455 W monocrystalline panels, coupled to a 12 kW Jua Power hybrid inverter and a high-capacity 16 kWh LiFePO4 battery with integrated BMS communication and automated transfer switching.

Key Technical Highlights:
- 7.28 kWp rooftop solar PV generation significantly slashes monthly utility grid electricity expenses
- 16 kWh LiFePO4 battery storage provides seamless backup power during grid blackouts
- 12 kW Jua Power hybrid inverter supports full domestic inductive loads and appliance spikes

Technologies: Jua Power 12 kW Inverter • 16 kWh LiFePO4 Battery • Rooftop Rail Mounting • DC Isolators & SPDs • Smart Energy Management • Wi-Fi Mobile Telemetry

09. BM Security Operations Center Solar Backup

C&I SOLAR & BESS | COMPLETED

Client: BM Security | Location: Nairobi, Kenya | Period: August 2026 | Contractor: Voltifix New Energy SEZ Limited

PV ARRAY (DC)	INVERTER (AC)	STORAGE (BESS)	INVERTER BRAND
3.64 kWp DC	6 kW AC	10 kWh LiFePO4 Battery	Jua Power

Engineering Summary:
Mission-critical solar backup power system engineered for BM Security's central operations room, featuring an 8-panel (3.64 kWp) array and 10 kWh LiFePO4 battery storage to guarantee uninterrupted power for radio and server equipment.

Operational Challenge: BM Security's 24/7 central alarm and radio dispatch center in Nairobi coordinates rapid armed response; even a 3-second blackout drops live radio channels, surveillance walls, and alarm servers.

Engineering Solution: Engineered an isolated critical-power topology with an 8-panel solar array (3.64 kWp), a 6 kW Jua Power pure sine wave hybrid inverter, 10 kWh LiFePO4 storage, and dual high-sensitivity SPDs.

Key Technical Highlights:
- Zero-interruption power changeover safeguarding mission-critical surveillance and communication racks
- Dedicated 10 kWh LiFePO4 storage bank delivering continuous control-room uptime
- Isolated electrical topology shielding sensitive radio equipment from line noise and voltage spikes

Technologies: Jua Power 6 kW Inverter • 10 kWh LiFePO4 Storage • Clean Sine Wave Inversion • High-Sensitivity SPDs • Zero-Latency Transfer • Critical Ops Isolation

10. Mariakani Coastal Residential Hybrid PV System

RESIDENTIAL HYBRID | COMPLETED

Client: Private Residence | Location: Mariakani, Kilifi County, Kenya | Period: May 2026 | Contractor: Voltifix New Energy SEZ Limited

PV ARRAY (DC)	INVERTER (AC)	STORAGE (BESS)	INVERTER BRAND
12.18 kWp DC	12 kW AC	32 kWh LiFePO4	Deye

Engineering Summary:
Coastal-engineered 12.18 kWp residential hybrid solar and 32 kWh LiFePO4 storage system in Mariakani, built to reduce overall power costs from the utility grid, eliminate power shortages during severe coastal grid disturbances, and withstand high ambient saline conditions.

Operational Challenge: High electricity bills from the utility grid and frequent, prolonged grid disturbances in Kilifi County caused frustrating power shortages during sweltering heat, while harsh saline humidity posed corrosion hazards to electrical hardware.

Engineering Solution: Installed 21 units of 580 W panels on coastal-grade anodized aluminum racking, paired with a 12 kW Deye hybrid inverter with peak-shaving control and 32 kWh modular Valley LiFePO4 batteries to maximize solar self-generation.

Key Technical Highlights:
- Substantially reduces overall power costs from the utility grid via high-output coastal solar array
- Minimizes power shortages during grid disturbances with 32 kWh modular LiFePO4 storage and auto-start
- Tailored for high ambient temperatures and saline coastal humidity with marine-grade fasteners

Technologies: Deye 12 kW Inverter • Valley Lithium Batteries • Peak-Shaving Control • Generator Auto-Start • Marine-Grade Racking • Earthing Continuity Verification

11. Willis Auto Spa Solar PV & Pumping Energy System

C&I SOLAR & BESS | COMPLETED

Client: Willis Auto Spa | Location: Kiambu County, Kenya | Period: July 2026 | Contractor: Voltifix New Energy SEZ Limited

PV ARRAY (DC)	INVERTER (AC)	STORAGE (BESS)	INVERTER BRAND
15.47 kWp DC	36 kW AC	32 kWh LiFePO4	Jua Power

Engineering Summary:
Solar PV and energy storage solution engineered to power high-pressure car wash pumps and commercial vehicle detailing bays with clean solar power and 32 kWh emergency battery backup.

Operational Challenge: High-pressure ceramic plunger pumps and vacuum extraction motors cycle intermittently hundreds of times daily, stalling operations and damaging switchgear during grid blackouts.

Engineering Solution: Deployed a 15.47 kWp rooftop solar array and three 12 kW Jua Power parallel hybrid inverters with 32 kWh LiFePO4 storage, fine-tuning inverter firmware surge thresholds for cyclic pump loads.

Key Technical Highlights:
- Programmed surge thresholds tailored to eliminate tripping during high-inrush pressure pump starts
- Weather-tight metallic conduit runs protecting high-stress commercial electrical pathways
- 32 kWh parallel LiFePO4 storage maintaining detailing operations during utility cuts

Technologies: Jua Power Hybrid Inverters • 32 kWh LiFePO4 Storage • Metallic Conduit Pathways • High-Pressure Pump Power • Inrush Surge Tuning • Earthing Loop Verification

12. Oaklands 5 Villas Residential Hybrid System

RESIDENTIAL HYBRID | COMPLETED

Client: Oaklands 5 Villas | Location: Kiambu County, Kenya | Period: August 2026 | Contractor: Voltifix New Energy SEZ Limited

PV ARRAY (DC)	INVERTER (AC)	STORAGE (BESS)	INVERTER BRAND
4.55 kWp DC	6 kW AC	16 kWh LiFePO4	Jua Power

Engineering Summary:
Standardized multi-villa residential solar and storage deployment at Oaklands 5 Villas, featuring a 4.55 kWp rooftop PV array, 6 kW Jua Power hybrid inverter, and 16 kWh LiFePO4 battery pack engineered to reduce overall power costs from the utility grid and safeguard homes against grid disturbances and power shortages.

Operational Challenge: High utility grid electricity bills and frequent localized grid disturbances subjected villa homeowners to recurring power shortages, creating a need for an estate-approved microgrid solution that complied with strict community aesthetic guidelines.

Engineering Solution: Standardized a 4.55 kWp rooftop solar pergola system, a 6 kW Jua Power hybrid inverter, and 16 kWh LiFePO4 battery storage with pre-engineered conduit pathways and aesthetic structural integration.

Key Technical Highlights:
- Reduces overall power costs from the utility grid via standardized rooftop pergola solar generation
- Safeguards villa homes against grid disturbances and power shortages with 16 kWh LiFePO4 storage
- Standardized scalable residential hybrid architecture suited for multi-unit gated villa communities

Technologies: Jua Power 6 kW Inverter • 16 kWh LiFePO4 Battery • Roof Truss Mounting • DC/AC Conduits • Pre-Commissioning Verification • Residential ATS

13. Power Analysis & Load Calculations

IOT & AUTOMATION | CONTINUOUS

Client: Residential & Industrial | Location: Nairobi & Regional Industrial Zones | Period: January 2025 – Present | Contractor: Engineering Consultancy / Technical Services

PV ARRAY (DC)	INVERTER (AC)	STORAGE (BESS)	INVERTER BRAND
Demand Profiling & Logging	Capacity Sizing Models	Phase Balancing & Harmonics	Power Quality Analyzers & Energy

Engineering Summary:
Conducted comprehensive electrical power quality analyses and dynamic load profiling across residential and industrial facilities. Translated operational demand curves and peak usage patterns into precise sizing models for hybrid solar PV, Battery Energy Storage Systems (BESS), and backup generation—ensuring power reliability, optimal power factor, and regulatory compliance.

Operational Challenge: Residential and industrial facilities frequently faced excessive utility peak kVA demand penalties, severe three-phase unbalance, and harmonic distortion that overheated transformers, overloaded neutral conductors, and caused equipment mis-sizing.

Engineering Solution: Conducted detailed power quality logging and dynamic demand profiling using advanced analyzers, formulating continuous and surge load models in Python and Excel alongside AutoCAD Electrical Single Line Diagrams (SLDs) to precisely size solar PV, BESS C-rates, and protection equipment.

Key Technical Highlights:
- Demand Profiling & Logging: Analyzed historical interval data, peak kVA demand, and power factor logs isolating critical vs non-critical loads
- Phase Balancing & Harmonic Mitigation: Evaluated three-phase unbalance and THD across distribution panels to eliminate neutral overloading
- System Capacity Sizing: Formulated continuous and surge calculations to appropriately size PV array capacities, inverter ratings, and battery C-rates

Technologies: Power Quality Analyzers • Energy Loggers • Python Load Modeling • Advanced Excel Modeling • AutoCAD Electrical • Single Line Diagrams (SLDs) • IEC 60364 • IEE

14. Tsuku Tsuku Hospitality Solar & Storage System

C&I SOLAR & BESS | COMPLETED

Client: Tsuku Tsuku | Location: Kiambu County, Kenya | Period: July 2026 | Contractor: Voltifix New Energy SEZ Limited

PV ARRAY (DC)	INVERTER (AC)	STORAGE (BESS)	INVERTER BRAND
30.03 kWp DC	36 kW AC	96 kWh LiFePO4	Jua Power

Engineering Summary:
Reliability-focused 30.03 kWp commercial solar installation delivering 36 kW AC combined three-phase capacity and a 96 kWh LiFePO4 high-autonomy battery bank for business continuity.

Operational Challenge: Commercial walk-in cold rooms and hospitality kitchens faced stock spoilage and revenue disruption during lengthy evening blackouts, with noisy auxiliary generators draining operational profits.

Engineering Solution: Installed a 30.03 kWp rooftop solar PV array driving three 12 kW Jua Power hybrid inverters in master-slave 3-phase parallel with a high-autonomy 96 kWh LiFePO4 battery bank.

Key Technical Highlights:
- 96 kWh high-autonomy LiFePO4 battery reservoir offering 14+ hours of critical load autonomy
- Synchronized 3-phase master-slave parallel inverter architecture
- Maintains strict phase symmetry across commercial refrigeration and processing equipment

Technologies: Jua Power Hybrid Inverters • 96 kWh LiFePO4 Storage • Master-Slave Parallel Operation • Busbar Distribution • Phase Symmetry Balancing • Circuit Isolators

15. ESP32 Cloud-Connected EV Charge-Point Status Reporter

IOT & AUTOMATION | COMPLETED

Client: Engineering R&D Infrastructure | Location: Nairobi, Kenya | Period: June 2026 | Contractor: Independent R&D

PV ARRAY (DC)	INVERTER (AC)	STORAGE (BESS)	INVERTER BRAND
Station Power Coupling	AC & DC EV Fast-Charger	Bidirectional MQTT Cloud	Embedded Controller

Engineering Summary:
Embedded hardware telemetry unit designed for AC & DC EV charging stations, utilizing FreeRTOS firmware in Embedded C and noise-isolated PCB circuitry to provide real-time status reporting, error alerting, and remote cloud commands.

Operational Challenge: EV charging stations required real-time cloud monitoring and remote control, but commercial controllers were costly and vulnerable to electrical noise from nearby high-current contactor switching.

Engineering Solution: Developed an embedded IoT telemetry unit around the ESP32 using FreeRTOS and Embedded C, featuring a noise-isolated 2-layer PCB with optocouplers and an MQTT state machine with local NVS caching.

Key Technical Highlights:
- Multi-tasking FreeRTOS firmware running non-blocking MQTT state machines
- Custom noise-isolated PCB layout built to withstand high-power AC & DC switching spikes
- Simulated and validated under Wokwi hardware-in-the-loop virtual environment

Technologies: ESP32 • FreeRTOS • ESP-IDF • Embedded C • MQTT State Machine • Non-Volatile Storage (NVS) • EasyEDA PCB Design • Wokwi Simulation

16. Rainbow Resort Commercial Hybrid Solar PV & High-Capacity BESS

C&I SOLAR & BESS | COMPLETED

Client: Rainbow Resort | Location: Kiambu County, Kenya | Period: June 2026 | Contractor: Voltifix New Energy SEZ Limited

PV ARRAY (DC)	INVERTER (AC)	STORAGE (BESS)	INVERTER BRAND
145.60 kWp DC	**150 kW AC**	**336 kWh LiFePO4**	**Jua Power**

Engineering Summary:
Engineered and deployed a high-capacity 145.60 kWp commercial hybrid solar PV and 336 kWh LiFePO4 battery storage system for Rainbow Resort, eliminating diesel generator reliance, maximizing self-consumption, and providing uninterrupted power during utility grid outages.

Operational Challenge: Volatile grid outages and crippling monthly diesel fuel costs threatened resort operations, while noisy generator runtime and voltage sags disrupted guest comfort.

Engineering Solution: Engineered a 145.6 kWp canopy solar array coupled to three 50 kW Jua Power parallel hybrid inverters and 336 kWh industrial LiFePO4 battery storage with automated transfer switchgear.

Key Technical Highlights:

- 336 kWh LiFePO4 battery storage across three 112 kWh industrial outdoor cabinets with automated combiner
- Suppressed diesel generator runtime during frequent utility load interruptions
- Synchronized 150 kW AC continuous capacity with dual Type I+II SPD electrical protection

Technologies: Jua Power Hybrid Inverters • 336 kWh LiFePO4 Cabinets • Master-Slave BMS Orchestration • Dual Type I+II SPDs • Automatic Transfer Switchgear • DC Shunt-Trip Disc

17. Industry 4.0 Real-Time Energy Telemetry & Cloud Analytics

IOT & AUTOMATION | COMPLETED

Client: Digital Qatalyst Projects | Location: Nairobi & Dubai | Period: October 2024 – May 2026 | Contractor: Digital Qatalyst

PV ARRAY (DC)	INVERTER (AC)	STORAGE (BESS)	INVERTER BRAND
Sub-Station Meter Network &	**Multi-Point Power Monitoring**	**Real-Time Cloud Dashboards &**	**N/A (Smart Meters)**

Engineering Summary:
Engineered an edge IoT energy telemetry system connecting commercial switchboard power meters via Raspberry Pi Linux gateways to MQTT and Dockerized Python analytics pipelines, achieving an 18% reduction in unmetered energy waste.

Operational Challenge: Commercial facilities operated blind to real-time power factor penalties, phase unbalance, and unmetered energy waste, leading to unexpected switchgear trips and high utility demand charges.

Engineering Solution: Deployed multi-channel RS485 Modbus RTU power meters linked to Raspberry Pi edge gateways running Dockerized Python microservices that stream electrical telemetry over MQTT to cloud dashboards.

Key Technical Highlights:

- Demonstrated actionable 18% energy waste reduction on monitored commercial panels
- Sub-second telemetry processing over lightweight MQTT brokers and Dockerized microservices
- Predictive maintenance alerts triggered by power-factor degradation and phase-unbalance thresholds

Technologies: RS485 Modbus RTU • MQTT Broker • Python Pipelines • Docker Microservices • Raspberry Pi Gateways • Time-Series Dashboards • Harmonics Analysis